Voice Deepfakes

Jacqueline Duong
UCLA
jduong2@g.ucla.edu

Alexander Edwards
UCLA
aledwards25@g.ucla.edu

Asia Khalid
UCLA
akhalid17@g.ucla.edu

Anika Lima
UCLA
anikalima@g.ucla.edu

Supavit Pitaknarongphorn
UCLA
supavit@g.ucla.edu

Abstract

This report investigates the detection and believability of voice deepfakes within immersive virtual reality (VR) environments by combining synthetic audio generation, spatial audio anchoring, visual avatar integration, and both human and machine evaluation. We generated deepfake audio using multiple models, including XTTS, and tested them against real voice samples in various conditions—such as with added background noise and directional sound placement. Participants in a custom VR environment were asked to distinguish between real and synthetic audio, while machine learning models, including RawNet2 and prosody-based detectors, evaluated the same samples. Results showed that advanced deepfakes, particularly those generated by XTTS, were more likely to deceive both humans and machines, though background noise and spatial context influenced detection accuracy. Technical limitations in avatar realism and synchronization presented challenges but also informed future work directions. This study provides a multi-modal foundation for understanding how deepfakes are perceived and detected in extended reality.

1. Introduction

As synthetic media technologies rapidly advance, the line between real and artificial content becomes increasingly blurred. Among these developments, audio deepfakes—AI-generated voice recordings that convincingly mimic real human speech—pose unique challenges in the realm of trust, authenticity, and security. While previous studies have primarily evaluated deepfake detection in 2D or passive listening contexts, few have explored how deepfakes are perceived within immersive extended reality (XR) environments, where spatial audio, visual avatars, and user interaction can significantly influence believability. This project aims to bridge that gap by investigating how synthetic speech is interpreted in VR settings and whether human and machine systems can reliably detect such content when integrated into realistic digital interactions.

To address these questions, we designed a series of experiments involving deepfake and real audio samples, spatial audio placement, visual avatar integration, and both human and machine detection methods. Our work explores multiple dimensions of perception—acoustic, visual, and spatial—and tests the effectiveness of current detection models, including both prosody-based classifiers and state-of-the-art machine learning systems such as RawNet2. Through user studies conducted in a custom-built VR environment, we examine how factors like noise, gender, and source direction affect a listener's ability to identify deepfakes. By combining technical implementation with perceptual analysis, this study provides a foundation for understanding the vulnerabilities and potential safeguards surrounding deepfakes in immersive media.

2. Background and Related Work

2.1 Prosodic Analysis

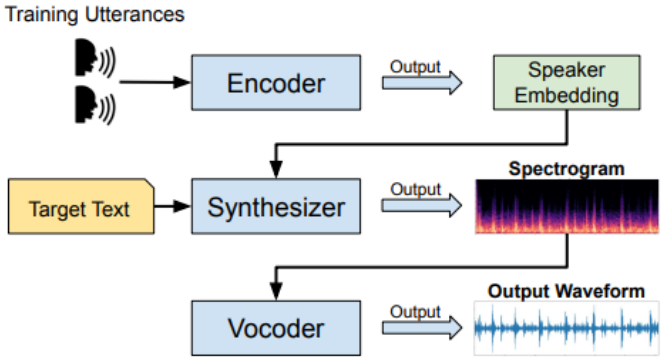


Figure 2.1.1: Previous work's illustration of the process for generating a deepfake audio. The encoder generates a voice embedding, the synthesizer creates a spectrogram for a target phrase, and the vocoder converts the spectrogram into a synthetic waveform.

"Pitch Imperfect: Detecting Audio Deepfakes Through Acoustic Prosodic Analysis"[1] proposes a novel approach to audio deepfake detection by focusing on prosodic features, which are high-level speech elements such as pitch, intonation, jitter, shimmer, and harmonic-to-noise ratio (HNR), that mirror how humans perceive natural speech. Unlike traditional detection models that rely on low-level spectral cues or black-box neural networks, their method targets the subtle vocal patterns that are most difficult for synthetic voices to replicate. Using the ASVspoof2021 dataset, they trained a prosody-based detector and achieved a 93% detection accuracy, demonstrating competitive performance with state-of-the-art machine learning models. More importantly, they highlighted the robustness of their approach against adaptive adversaries and showcased how attention mechanisms could explain which features had the

greatest influence on the model's decisions, namely, jitter, shimmer, and mean pitch (F0).

This work is particularly relevant to our research, as it underscores the importance of prosody in both detecting deepfakes and understanding how humans interpret synthetic speech. It also provides a valuable framework for evaluating the realism of generated audio, especially in the context of immersive environments where intonation, emotion, and cadence play crucial roles in perception. By referencing this prosody-based detection method, we aim to complement their insights with new dimensions of analysis, such as spatial audio placement and multimodal avatar consistency in XR. Our project builds on their foundational concept that audio realism is not merely technical, it is perceptual, and that detection strategies must evolve alongside the environments in which deepfakes are deployed.

2.2 Deepfake Evaluation

The "Better Be Computer or I'm Dumb": A Large-Scale Evaluation of Humans as Audio Deepfake Detectors"[2] study presents the largest known user-centered evaluation of audio deepfakes, involving over 1,200 participants and thousands of audio clips from top datasets including ASVspoof2021, WaveFake, and FakeAVCeleb. The authors found that human listeners generally outperform machine learning models in detecting real human audio but struggle more with detecting deepfakes. Importantly, the study revealed that human decisions were strongly influenced by prosodic and linguistic cues such as tone, cadence, background noise, and natural breathing, factors often ignored by automated detectors. Conversely, machine models had higher false positive rates, often misclassifying natural audio as synthetic, especially when quality was low or voice patterns were atypical.

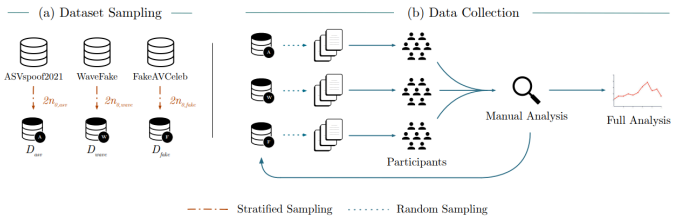


Figure 2.2.1: Previous work's methodology diagram. They extracted data samples, had them listened to by participants, discarded bad responses, and collected data.

This work directly informs our research approach by emphasizing the role of user perception in deepfake detection, particularly in environments that demand real-time interaction and immersion. The study's findings validate the need for human-in-the-loop evaluation and expose the limitations of relying solely on machine detection, especially when deepfakes aim to deceive humans, not machines. Our work extends their findings into the realm of spatial XR, where voices are coupled with visual and locational cues, and where trust may hinge not just on sound quality but also on contextual alignment. We adopt their survey-driven methodology to assess believability while layering in spatial realism, thus pushing the boundary of where and how audio deepfakes can be detected or go unnoticed.

3. Threat Model

3.1 Adversarial Goals

The primary objective of a voice deepfake adversary is to generate synthetic speech that is indistinguishable from real human voices, thereby deceiving either human users, automated systems, or both. In our context, the attacker aims to exploit trust in immersive XR settings by introducing realistic synthetic avatars with convincing voices. These attacks may be used for impersonation, misinformation, social engineering, or to undermine authentication systems. The adversary's success is measured by their ability to influence perception, decisions, or actions based on the synthetic voice content.

3.2 Capabilities of the Adversary

We assume a moderately powerful adversary with access to modern open-source TTS models such as XTTS-v2 or Kokoro, and the computational resources to generate high-fidelity, prosodically-rich synthetic speech. The adversary may possess voice samples of a target speaker (for cloning), textual scripts (for voice synthesis), and even the ability to embed these voices within virtual avatars in XR platforms. However, we do not assume access to user biometric data or proprietary detection systems. The adversary may optionally apply adversarial perturbations to evade detection systems or manipulate prosodic features to improve realism.

3.3 Attack Surface in XR

Extended reality platforms introduce a unique and expanded attack surface for voice deepfakes. Unlike traditional media, XR combines spatial audio, visual presence, and interactive behaviors, making it a fertile ground for multimodal deception. Voice deepfakes can be delivered via avatars in multiplayer VR sessions, embedded in NPCs (non-player characters), or used during virtual meetings to impersonate colleagues. The immersive nature of XR can reduce skepticism, especially when synthetic voices are aligned with realistic avatars and spatial cues, increasing the effectiveness of attacks and complicating detection.

3.4 Targeted vs. Untargeted Threats

Voice deepfake attacks can be targeted, impersonating specific individuals such as a known speaker or public figure, or untargeted, aiming to sound generally human to exploit trust in anonymous interactions. Targeted attacks require voice cloning and often have more serious implications (e.g., fraud, manipulation), while untargeted attacks might be used for spam, influence operations, or psychological disruption in XR environments. Our threat model accounts for both, but emphasizes targeted threats given their higher stakes in XR's socially interactive settings.

3.5 Detection Challenges and Threat Persistence

In XR environments, traditional detection methods face new challenges due to real-time interaction, spatial dispersion of audio sources, and the integration of voice with visual and behavioral cues. The adversary benefits from these multimodal distractions, as users may focus more on avatar expression or environmental context than audio scrutiny. Additionally, many existing detectors are trained on isolated audio samples, not spatialized or embedded voices. This gap allows deepfake threats to persist longer in XR, where detection tools and user instincts may not yet be sufficiently developed or aligned.

4. Overview Of The Design

Our system is designed to evaluate the realism and detectability of voice deepfakes within an immersive extended reality (XR) environment by combining synthetic audio, spatial audio placement, and avatar interaction. We begin by generating deepfake speech using three models, ASVspoof2021, Kokoro, and XTTS-v2, each varying in fidelity, prosody, and speaker specificity. These audio clips are then integrated into a Unity-based VR scene that includes spatial audio anchors, a visual avatar, and user interface elements for interaction and voting. Participants enter the environment, listen to each voice from different spatial positions, and vote on whether the audio is human or computer-generated. The design facilitates both subjective human evaluation and compatibility testing with machine detection tools, allowing us to study how audio realism and spatial immersion affect users' ability to discern synthetic speech.

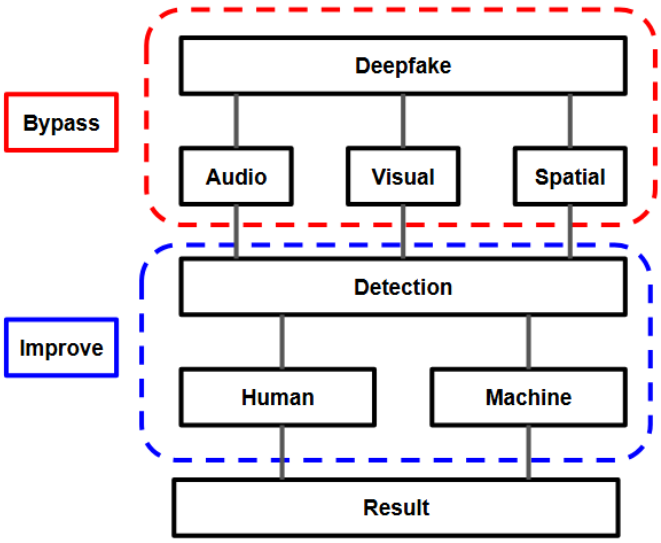


Figure 4.1: Diagram for the design. We start with developing deepfakes with the goal of bypassing human and machine detection using audio, visual, and spatial elements. Then, we will move on to improving human and machine detection capabilities, either through education, training, or algorithms.

5. Methodology/Design

5.1 Audio Deepfake

To generate high-quality synthetic speech for evaluation, we employed three different audio deepfake models: ASVspoof2021, Kokoro, and XTTS-v2. The ASVspoof2021 dataset provided short, pre-generated clips commonly used in detection research, although many included background noise and were less suitable for immersive applications. Kokoro was selected for its efficiency and ability to produce longer, prosodically rich audio in various accents and emotional tones. XTTS-v2, by contrast, was used for its advanced voice cloning capabilities, enabling us to replicate specific individuals' voices with high fidelity. For controlled comparison, we generated audio clips from the same scripts across models and speakers, including male, female, and accented voices. This consistency allowed us to analyze prosodic characteristics such as pitch, cadence, and intonation, and to test how different synthesis approaches impacted believability and detection within the XR environment.

Later, we gathered more datasets comprising both genuine human-recorded and deepfake audio clips. To test the impact of environmental noise, we augmented both real and synthetic clips by layering background sounds, including ambient chatter and white noise. This allowed us to evaluate whether such noise would increase the perceived believability of deepfakes by masking imperfections in prosody or voice synthesis. These variations were included in user experiments to assess how background noise affects detection accuracy and user confidence, particularly in immersive XR settings where such auditory interference is common.

5.2 Visual Deepfake

For the visual component of our deepfake system, we integrated animated avatars into the XR environment to match and enhance the realism of the synthetic audio. Using Unity as the development platform, we created virtual avatars capable of head and lip movement synchronized with the audio playback. While we did not generate fully synthetic facial deepfakes, we focused on ensuring multimodal consistency between voice and visual cues, aligning the avatar's gender, accent, and speaking style with the characteristics of the corresponding deepfake audio. Spatial audio anchors were positioned to reflect the avatar's location in 3D space, enhancing the illusion of presence. This visual integration was critical to evaluating how contextual cues, such as avatar realism, gaze, and spatial alignment, influence a user's ability to detect audio deepfakes. By embedding the voices within believable digital personas, we created a more realistic and immersive testbed to assess trust and deception in multimodal communication.

During development, we explored the use of Unreal Engine's Metahuman avatars, which offered significantly more realistic facial expressions and lip-syncing capabilities. However, because our XR controller logic was built in Unity, we encountered incompatibility issues that prevented full integration of the Metahuman system. As a result, we used ReadyPlayerMe avatars in Unity, which offered basic support for lip-syncing. Overcoming these limitations required hours of trial-and-error adjustments, as there was a lack of

up-to-date tutorials and documentation for achieving high-quality voice-to-avatar synchronization within Unity.

5.3 Spatial Deepfake

To simulate realistic interactions in an immersive environment, we implemented spatial audio techniques that positioned voice deepfakes within a 3D XR space. Using Unity's built-in spatial audio system, we anchored each synthetic voice to a specific location in the virtual environment, allowing participants to perceive directional sound cues as they moved around. This spatialization was designed to match the position of the visual avatar, ensuring that audio appeared to originate naturally from the avatar's mouth and body orientation. By doing so, we explored how spatial realism influences the perceived authenticity of deepfake speech. We hypothesized that spatial audio, by grounding the voice in a believable physical source, could enhance user trust and reduce scrutiny. Participants engaged with the environment by navigating between avatars, listening to their speech from various angles, and then classifying each voice as human or computer-generated. This setup allowed us to assess how spatial context affects the effectiveness and detectability of voice deepfakes in dynamic XR settings.

Building on this spatial framework, we specifically designed one experiment to test whether the direction from which a voice deepfake is heard, left versus right, affects its believability. The VR environment was configured so that participants listened to audio samples while oriented to either side, with voice sources spatially anchored to consistent left or right positions. We then conducted user experiments to gather empirical responses on detection accuracy under these spatial conditions. This directional test allowed us to isolate and evaluate the cognitive effects of audio positioning, contributing deeper insight into how spatial orientation influences user perception of synthetic versus real speech.

5.4 Human Detection

To evaluate how well users can detect voice deepfakes, we conducted a human subject experiment with .wav files we generated from deepfake models. Twelve participants were recruited to listen to the audio files, which were either human-recorded or synthetic speech. Participants were able to play the files, then classify each voice as either "Human" or "Computer-Generated." Each session presented a randomized mix of audio clips generated by ASVspoof2021, Kokoro, and XTTS-v2, alongside genuine human samples. After completing all classifications, participants answered a short survey asking what cues influenced their decisions, such as tone, cadence, or emotional expressiveness. This setup allowed us to collect both quantitative detection accuracy and qualitative reasoning behind participant judgments. The experiment provided insights into how effectively voice deepfakes deceive users in audio environments.

In a later phase of the study, we expanded our human detection experiments by involving more participants and introducing a broader range of test conditions. The audio samples were diversified to include additional voices and deepfakes with varying degrees of prosodic realism. Moreover, we introduced visual, audio, and spatial manipulations—such as avatar alignment, background noise, and directional audio placement—to examine how these factors influenced user detection performance. By comparing detection accuracy across these manipulated variables, we systematically tested hypotheses related to believability and deception, providing a more comprehensive understanding of which multimodal features most strongly affect human perception of synthetic speech in XR settings.

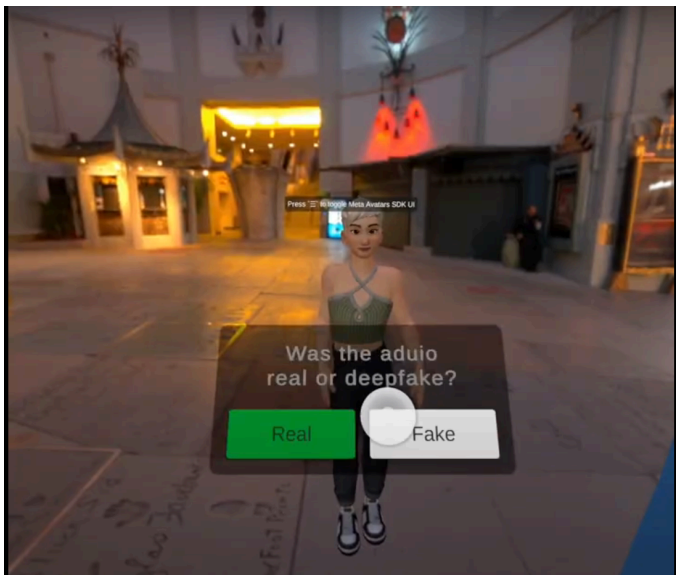


Figure 5.4.1: This illustrates the environment in which participants experience during the experiment, including the position of the avatar, sound, and how they give responses.

5.5 Machine Detection

To complement human evaluation, we tested the generated voice samples against an automated machine detection model based on prosodic analysis. Using the methodology outlined in Pitch Imperfect, we extracted six key prosodic features from each audio clip: mean and standard deviation of fundamental frequency (F0), jitter, shimmer, and harmonic-to-noise ratio (HNR), using the Parselmouth Python interface for Praat. These features were then fed into a trained classifier to determine whether each sample was human or synthetic. Our goal was to assess whether machine detectors could identify deepfakes that successfully fooled human listeners, particularly those generated by advanced models like XTTS-v2. We also analyzed discrepancies between human and machine classifications to identify patterns, such as whether machines were more sensitive to prosodic flaws that humans overlooked, or vice versa. This dual evaluation enabled us to benchmark detection performance across both human and machine perspectives, highlighting the strengths and blind spots of each.

In addition, we utilized a pretrained model from the ASVspoof2021 challenge, which had been trained on the ASVspoof2019 dataset, to further evaluate machine detection capabilities[6]. This model allowed us to analyze how state-of-the-art detection systems perform on our generated

audio, particularly under new test conditions. We categorized the model's classification results by various attributes, such as speaker gender, presence of background noise, and source model, to determine if any specific attribute consistently influenced classification accuracy or confidence. This attribute-based breakdown provided deeper insights into the types of audio manipulations that may either aid or hinder automated detection systems.

6. Preliminary Results

6.1 Audio Deepfake

Our preliminary results show clear differences in the effectiveness of various audio deepfake models based on both human perception and technical quality. We conducted a survey with 12 participants who were asked to classify a set of audio clips, some human, others generated by ASVspoof2021, Kokoro, and XTTS-v2 models, as either real or computer-generated. The results revealed that participants were generally more accurate at identifying real human speech, with 87.5% correctly classifying genuine clips. XTTS-v2 deepfakes were the most successful at deceiving listeners, with only 70.8% of synthetic samples correctly identified, likely due to their voice cloning capabilities, which replicate speaker-specific prosodic traits such as tone and cadence. In contrast, Kokoro-generated audio was correctly identified as fake by 100% of participants, suggesting that, despite its support for emotional and cross-lingual synthesis, its outputs lacked the subtle variations in inflection that characterize natural speech. The ASVspoof2021 dataset fell in between, while most participants identified its samples as synthetic, some found its shorter clips harder to evaluate confidently. Participants frequently cited tone, inflection, and cadence as key cues in making their decisions. These findings emphasize the importance of prosodic features in human deepfake detection and provide valuable insights into which aspects of synthetic speech are most effective at mimicking human audio.

6.2 Visual Deepfake

We experimented with realistic avatars to complement the synthetic speech used in our XR environment. The goal of these avatars was to design them to align visually with the voice characteristics, such as gender, accent, and speaking style, to ensure consistency between audio and visual modalities. Our development efforts focused on implementing synchronized head and mouth movements, natural posture, and expressive behaviors that matched the tone and rhythm of the deepfake audio. This visual coherence was critical for our upcoming experiments, as we aimed to evaluate how avatar realism and animation influenced the perceived authenticity of voice deepfakes. The integration of these elements allowed us to test whether a convincing visual presentation could enhance or diminish users' ability to detect synthetic speech in immersive settings.

During the development process, we experimented with Unreal Engine's Metahuman avatars, which offered superior visual realism and mouth movement accuracy but were ultimately incompatible with our Unity-based application logic. Due to this constraint, we used ReadyPlayerMe avatars within Unity, which supported only a basic WAV-driven lip-sync system. Achieving realistic mouth movement required extensive manual adjustments, including fine-tuning of teeth and lip synchronization, and ensuring that the audio speaker instance was precisely placed in the scene hierarchy to prevent delays. The lack of up-to-date tutorials made this process heavily reliant on trial-and-error troubleshooting. Despite these limitations, we successfully developed a working solution in Unity that synchronized audio playback with visible mouth movement, enabling us to conduct our experiments with integrated visual and auditory deepfakes. While the solution was sufficient for the experiment to function, there was considerable room for improvement in making it more realistic and believable.

6.3 Spatial Deepfake

We completed the development of the spatial deepfake environment within Unity, setting up foundational components such as spatial audio anchors, directional sound playback, and avatar-linked positioning. Our setup allowed synthetic voices to be projected from distinct locations in the 3D environment, creating the groundwork for immersive audio-visual interactions. We worked on integrating avatar animations, refining spatial synchronization, and implementing experimental conditions to test how spatial realism influenced the believability of deepfakes. Once finalized, this environment enabled us to evaluate how spatial audio affected user perception and whether spatial context could amplify or mitigate the effectiveness of synthetic speech in XR.

Experimental Setup - Part 1:

Does listening to a deepfake from the left or right side affect its believability?

	Turned Left	Turned Right
% Accuracy	69.2%	68.3%

Figure 6.3.1: The results from the spatial deepfake experiment, displaying the accuracy of participants hearing audio without any manipulation from the left versus the right side.

After completing the spatial audio setup, we conducted an experiment to assess whether the direction of audio presentation—left vs. right—impacted the believability of deepfake speech. Ten participants were recruited and split evenly: five listened to audio while turned left, and five while turned right. Each participant was exposed to a randomized set of 24 audio samples, comprising 12 real and 12 deepfake clips, all without background noise. The resulting

classification accuracy was 69.2% for those turned left and 68.3% for those turned right, indicating that spatial orientation had minimal effect on users' ability to distinguish between real and synthetic speech. These findings suggested that in the absence of conflicting visual or environmental cues, directional audio alone did not significantly influence believability.

6.4 Human Detection

In a follow-up experiment, we examined whether the presence of background noise, specifically ambient chatter or white noise, affected participants' ability to distinguish between real and deepfake audio. Another ten participants, evenly split between left- and right-facing orientations, were presented with 24 audio clips (12 real, 12 synthetic) embedded with background sounds. The results showed that participants performed slightly better under noisy conditions, with accuracies of 77.5% (turned left) and 76.7% (turned right). Interestingly, deepfakes without background noise were misclassified 30% of the time, compared to just 22.5% when noise was present. Likewise, real audio was misclassified more frequently without noise (32.5%) than with it (23.3%). These findings suggest that background noise can aid human listeners in detecting deepfakes, possibly by making unnatural vocal patterns more apparent.

Experimental Setup - Part 2:

Does adding background noise to a deepfake increase its believability?

	Turned Left	Turned Right
% Accuracy	77.5%	76.7%

Figure 6.4.1: The results from the human detection experiment, displaying the accuracy of participants hearing audio with background noise from the left versus the right side.

To evaluate whether immersion in a VR environment affects deepfake detection accuracy, we compared our experimental results against data reported in prior non-VR studies. Analysis of classification accuracy revealed that participants were able to distinguish between deepfake and real audio in the VR setup with similar effectiveness to traditional listening setups. This was evident in our visual results, where the two rightmost columns of the comparison graph—representing the "no noise" and "noise" conditions from our VR experiments—showed classification rates that closely matched the earlier, non-VR benchmarks depicted in the left columns. The left graph details accuracy per audio type (real vs. synthetic), while the right graph summarizes overall user accuracy across conditions. These findings suggest that spatial immersion in VR does not inherently reduce human capacity to detect synthetic speech.

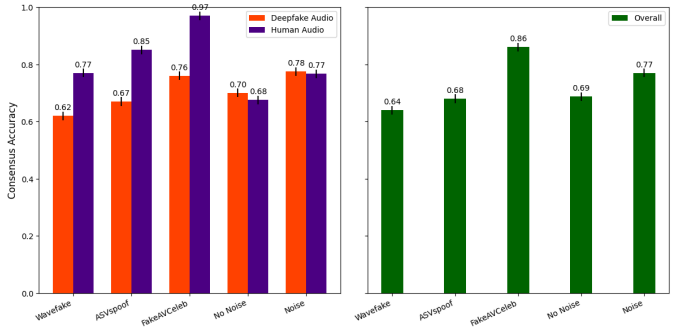


Figure 6.4.2: The left graph shows human detection accuracy for deepfake and human audio. Wavefake, ASVspoof, and FakeAVCeleb columns are from previous work conducted outside of the VR space. No noise and Noise columns are from our work conducted inside the VR space. The right graph shows average accuracy for each column.

6.5 Machine Detection

For the machine detection component, we began analyzing our generated audio clips using a prosody-based detection model inspired by the Pitch Imperfect approach. By extracting key prosodic features, mean and standard deviation of fundamental frequency (F0), jitter, shimmer, and harmonic-to-noise ratio (HNR), we aimed to evaluate how reliably the model could distinguish between human and synthetic speech. Initial tests showed that the model performed well on detecting ASVspoof2021 samples, which contain detectable artifacts and prosodic inconsistencies. However, the model exhibited reduced confidence when evaluating XTTS-v2 audio, likely due to its more natural-sounding pitch variation and speaker-specific cloning. These early results suggest that while traditional prosodic features are useful for detecting simpler deepfakes, more advanced models like XTTS-v2 may require enhanced or multi-modal detection techniques. Further analysis and dataset expansion are underway to validate these trends and improve classification robustness.

Machine Detector

	Real		Deepfake	
	Male	Female	Male	Female
Normal	-10.039	-2.498	-9.083	-6.916
Chatter	-11.211	-5.826	-10.094	-8.836
Noise	-8.212	-0.001	-9.356	-5.990

Figure 6.5.1: Results from the machine detector experiment organized by different attributes. If the result is closer to one, the detector is more confident that the audio is human. If the result is more negative, the detector is more confident that it is a deepfake.

To supplement our prosodic analysis, we evaluated all 36 audio samples using the RawNet2 baseline model from the

ASVspoof2021 Deepfake track, which was trained on the ASVspoof2019 dataset. The model output log-likelihood scores for each audio, stratified by gender (male, female), audio type (real, deepfake), and condition (normal, chatter, noise). Results showed that deepfake samples generally had lower scores (more negative), indicating higher confidence in detection, particularly for female voices and noise conditions. When compared to the prosody-based model from the Pitch Imperfect study, which achieved 93% accuracy and robustness against basic adversarial techniques, our model demonstrated weaker overall performance on our dataset.

7. Next Steps

7.1 Limitations

7.1.1 VR Environment

One of the key limitations we encountered during development was the difficulty of setting up a stable and immersive VR environment, particularly with regard to implementing accurate spatial audio. Anchoring audio to specific locations in 3D space required deep familiarity with Unity's spatial sound systems and careful configuration of audio sources, listener positions, and object hierarchies. Even small misalignments often cause the voice to appear disconnected from its visual source, diminishing immersion and breaking the illusion of presence. Fine-tuning spatial audio to match user perspective and avatar positioning was a time-consuming process and often lacked adequate documentation or support, making iterative adjustments labor-intensive and prone to error.

7.1.2 Visual Deepfake

Constructing realistic avatars that convincingly matched the speech audio presented additional challenges. While tools like ReadyPlayerMe offered basic character models, they lacked advanced facial animation capabilities needed to simulate expressive speech, hand gestures, and micro-expressions. The lip-sync system supported only one viable method for syncing .wav files, and achieving even moderate realism required extensive manual tweaking of teeth and lip movements. Moreover, syncing pre-recorded audio with mouth motion was highly sensitive to hierarchy placement within Unity, and any misconfiguration led to noticeable desynchronization. These issues highlight the need for more robust avatar systems and real-time voice-to-animation pipelines in future work.

7.2 Future Work

7.2.1 Visual Deepfake

A key direction for future work is enhancing avatar behavior to more closely resemble natural human interaction within a VR room. This includes implementing more lifelike gestures, facial expressions, gaze direction, and responsive body language that adapt dynamically to the flow of conversation. Achieving this level of realism would significantly improve the overall believability of the XR experience, helping to maintain user immersion and making deepfakes more difficult—or perhaps easier—to detect depending on their quality. Leveraging more advanced avatar systems or real-time animation pipelines could help bridge the gap between synthetic audio and visual behavior.

7.2.2 Dataset

To strengthen the generalizability and robustness of our findings, future efforts will focus on collecting a larger and more diverse set of audio samples. This includes deepfakes generated from various speakers, accents, and emotional tones, as well as more human-recorded baselines. With this expanded dataset, we plan to conduct additional user studies with a broader participant pool to explore detection accuracy across different demographics and listening conditions. These experiments would allow us to better understand the variability in human perception of deepfakes and validate findings across more realistic use cases.

7.2.3 Machine Detection

Another important area of future work involves testing our audio files against a wider range of machine detectors beyond RawNet2 and prosody-based models. This includes evaluating commercial-grade or open-source deepfake detectors and observing their vulnerability to common and novel manipulations. Additionally, we aim to experiment with adversarial techniques—such as subtle prosodic or spectral perturbations—to attempt to bypass these detectors. Insights gained from these tests can then be used to propose improvements, potentially leading to more robust, context-aware detection systems for XR environments.

7.2.4 Experiment

We also propose a new experiment in which multiple avatars interact in a shared VR chatroom, all using real human audio except for one that speaks with a deepfake voice. Participants would be tasked with identifying which avatar is synthetic based solely on their speech patterns, facial expressions, and spatial behavior. This setup introduces a more social and dynamic detection challenge, mirroring real-world scenarios where deepfakes might be embedded within group conversations. The goal is to explore how well users can detect synthetic voices in a realistic, interactive setting where contextual and behavioral cues play a larger role in perception.

8. Conclusion

This project explored the detectability and believability of voice deepfakes in immersive XR environments by combining synthetic audio, spatial sound anchoring, visual avatars, and both human and machine evaluation techniques. Our experiments revealed key insights into how different deepfake generation models are perceived, how spatial orientation and background noise affect human detection accuracy, and how current machine detection methods perform against high-fidelity synthetic speech. Despite technical limitations in avatar realism and synchronization, the integration of multimodal cues provided a robust platform for studying perception in extended reality. The findings underscore the

importance of holistic, context-aware approaches to deepfake detection, particularly as XR applications grow in sophistication and adoption. Our work lays the foundation for future research on multimodal deception and detection in interactive digital spaces.

9. Contributions Of Each Teammate

9.1 Jacqueline Duong

Jacqueline Duong contributed to the audio analysis component of the project by experimenting with various deepfake generation tools, including Bark from Hugging Face. She also conducted frequency domain analysis on both real and synthetic .wav files to identify potential distinguishing features between human and AI-generated speech. By normalizing and plotting the data, she assessed whether spectral differences could reveal inconsistencies.

Jacqueline collaborated closely with Anika to generate new deepfakes using the XTTS model and to collect real human audio samples for comparative analysis. Together, they systematically added background chatter and white noise to both synthetic and real audio clips to evaluate their effect on user detection accuracy—an effort central to testing the project's secondary hypothesis regarding the impact of noise on believability. Jacqueline also played an active role in conducting user studies within the VR environment developed by Alex, helping facilitate participant sessions and collect detection data under various spatial and auditory conditions.

9.2 Alexander Edwards

Alexander Edwards led the development of the XR environment by building the foundational spatial audio testbed in Unity, enabling the team to embed voice deepfakes into an interactive 3D space. His work included designing a user-friendly interface to start the application, placing spatial audio anchors to simulate directional sound, and implementing the logic for audio playback. He also created a voting interface for users to classify audio clips as real or fake, along with a dynamic scoreboard and a restart button to support multiple test sessions. These features form the core infrastructure for upcoming human detection experiments.

In addition to his initial development of the XR environment and user interface, Alexander Edwards was also responsible for creating and refining the full VR environment used in the final user experiments. He implemented spatial audio anchoring to ensure voices originated from the correct avatar positions, enabling realistic directional sound perception. Alexander also integrated the avatars into the scene, configured their interaction logic, and ensured seamless transitions between test conditions. His work on synchronizing user interactions, audio playback, and avatar behaviors was essential in delivering a smooth, immersive experience for participants, directly supporting the experimental goals of the project.

9.3 Asia Khalid

Asia Khalid led the development of the visual deepfake component of the project, focusing on creating realistic avatars that visually align with the characteristics of the synthetic audio. Her work involved designing and preparing 3D avatars with features such as gender, facial structure, and expression that match the corresponding voice samples. She also began implementing mouth movement and head animations to eventually synchronize with the timing and emotion of the audio, laying the foundation for believable multimodal interaction in XR. Asia’s efforts are related to ensuring that the visual presentation reinforces or challenges the authenticity of the voice, a critical factor in evaluating how deepfakes are perceived in immersive environments.

Asia Khalid dedicated significant time during the early stages of the project to independently researching available tools and evaluating potential workflows for achieving high-fidelity facial animation. She explored the capabilities of Unreal Engine's Metahuman avatars, including their advanced visual realism and lip-sync baking pipeline, and assessed the feasibility of integrating these assets into the team’s Unity-based framework. Asia also investigated the technical requirements for real-time deepfake voice-to-avatar synchronization and identified current limitations in existing systems. Despite extensive effort and several working prototypes, the visual avatar component ultimately could not be integrated into the final VR environment due to compatibility constraints and unresolved synchronization issues.

9.4 Anika Lima

Anika Lima was responsible for generating and curating the core dataset of voice deepfakes used in the project’s initial evaluation. She produced 12 deepfake audio clips using the Kokoro model, known for its prosody imitation and support for long, clean, cross-lingual speech synthesis, and 8 additional clips using XTTS-v2, which offers advanced voice cloning capabilities tailored to specific speakers. To ensure consistency across evaluations, Anika standardized the generated clips by using the same script across different models, incorporating both male and female voices and non-American accents to assess how prosodic variations affect realism. She also compared these generated samples with spoofed clips from the ASVspoof2021 dataset and conducted a human survey with 12 participants, asking them to classify the clips as real or computer-generated. The results provided key insights into which models were most effective at deceiving listeners.

Anika concentrated heavily on designing and executing the user experimentation component of the project. She collaborated with Jacqueline to generate new deepfakes using XTTS and collect real audio samples, which were then modified with background noise to support the investigation of the team’s secondary hypothesis on believability. Anika led the planning and facilitation of user studies within the VR environment, ensuring that participants experienced the audio under consistent spatial and auditory conditions. She was responsible for guiding users through the experimental procedure, collecting their responses, and managing data

quality throughout the testing process. Her focus on rigorous user testing helped ensure the reliability of the study's results.

9.5 Supavit Pitaknarongphorn

Supavit Pitaknarongphorn was responsible for the machine detection component of the project, focusing on evaluating whether synthetic audio could be reliably identified using automated techniques. Drawing from the Pitch Imperfect paper, he implemented a prosody-based detection pipeline that extracted key speech features such as mean and standard deviation of F0, jitter, shimmer, and harmonic-to-noise ratio (HNR) using the Parselmouth interface for Praat. He tested the generated deepfakes, XTTS-v2, Kokoro, and ASVspoof2021, against this model to assess its ability to distinguish synthetic from real speech. Early results showed strong detection of simpler deepfakes but reduced accuracy on more advanced audio like XTTS-v2, highlighting the need for more robust models.

Supavit was responsible for setting up and validating the detection workflow, ensuring compatibility between the extracted audio features and the trained models. Supavit ran the detection passes using the pretrained AVSspoof2021 model, using RawNet2 baseline, to evaluate their effectiveness on real versus deepfake audio under different noise and spatial conditions. His efforts ensured that the machine learning evaluation component was fully functional and could be directly applied to the experimental audio samples, generating critical insights for comparing machine and human detection performance.

Project Code: https://tinyurl.com/188projcode

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